MIL-C-46380(MU)
1 May 1963

MILITARY SPECIFICATION

CASES, COMBINATION, SET, W/WORK TABLES: 8625640

1. SCOPE

1.1 This specification covers a set of multipurpose combination cases with work tables and van mounting frame for DC, Low Frequency and Physical Calibration Standards. Combination cases are suitable for all types of shipment and storage. The set when installed in an M292 van is intended for field calibration functions. The set may also be used for calibration activities at fixed military installations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods Of
MIL-W-6858	Welding, Aluminum, Magnesium, Non-Hardening Steels or Alloys, and Titanium, Spot, Seam and Stitch
MIL-W-8604	Welding Aluminum Alloys, Process For
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-I-26860	Indicator, Humidity, Plug, Color Change
MIL-I-45208	Inspection Requirements - General Inspection For

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking for U. S. Military Property

FSC 4931

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DRAWINGS

Munitions Command

A8625640 Combination of Adopted Items (Set, Combination Cases and Work Tables for DC, Low Frequency and Physical Calibration Standards)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Combination cases, work table assemblies, van frame assembly and RF Shielding Kits shall be fabricated in accordance with the requirements specified herein. The set consists of the following:

QTY	NOMENCLATURE	DESIGNATOR OR DETAIL	PART NUMBER
1	Container Assembly	A-1	8625624
1	Container Assembly	A-2	8625625
1	Container Assembly	A-3	8625626
1	Container Assembly	A-4	8625627
1	Container Assembly	A-5	8625628
1	Container Assembly	B-6	8625629
1	Container Assembly	B-7	8625630
1	Container Assembly	B-8	8625631
1	Container Assembly	A-9	8625632
1	Container Assembly	A-10	8625633
1	Container Assembly	A-11	8625634-1
1	Container Assembly	A-12	8625634-2
1	Container Assembly	A-13	8625634-3
1	Container Assembly	A-14	8625634-4
1	Container Assembly	A-15	8625635-1
1	Container Assembly	A-16	8625635-2
1	Container Assembly	A-17	8625634-5
1	Container Assembly	A-18	8625635-3
1	Work Table Assembly		8625636-1
1	Work Table Assembly		8625636-2
1	Van Frame Assembly		8625637
1	Kit, R.F. Shielding and adapters		8625638

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3.1.1 Preproduction cases.- Prior to production, combination cases A-1, A-3, A-4 and B-7 shall be submitted for inspection and testing under Government surveillance. Inspection and testing shall be performed by the contractor using Government approved facilities. The cases shall be subjected to inspection and testing requirements specified in Section 4. Cases submitted for tests shall not be prime coated or finished.

3.1.2 Preproduction set.- A preproduction set consisting of items defined in paragraph 3.1 shall be assembled and van mounted by the contractor in an M292 van. The installation of the frame assembly and cases shall be in accordance with "Operating Instruction Manual" to be furnished with each set by the Government. The assembled preproduction set shall be capable of withstanding a road test as specified in Section 4.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification. Unless otherwise specified, the contractor shall provide certification of compliance to material specifications and processes.

3.3 Construction.- The entire set shall be manufactured in accordance with drawing A8625640 and all drawings and specifications pertaining thereto.

3.3.1 Welding.- Welds shall be reasonably smooth and free of craters. They shall exhibit characteristics of fusion and soundness of good welding practice and shall meet the requirements of Specification MIL-W-8604. Parts to be welded shall be clean, free from rust, scale, corrosion, oil and other foreign substances before welding operations are performed.

3.3.1.1 Spot welding.- Component parts assembled or attached by spot welding shall conform to applicable requirements of Specification MIL-W-6858.

3.3.2 Closure.- Case closure shall be as specified on applicable drawings. The gaskets shall be properly secured with no excessive adhesive remaining in closure joint.

3.3.2.1 Latch.- The latches shall be assembled to cases as required by applicable drawings. The latches shall be uniformly adjusted to maintain a positive pressure on the gasketed seal and shall not permit leakage when the case is subjected to pressure retention tests specified in 4.5.2.2.

3.3.3 Power receptacles.- Cases requiring power receptacle assemblies shall be wired and assembled to assure good electrical contacts and continuity. Grounding provision shall be provided from suspended frame to case shell.

3.3.4 Humidity indicator.- A humidity indicator plug conforming to Type II of Specification MIL-I-26860, shall be installed adjacent to the breather valve in each case. (see 6.1.2)

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3.3.5 Cushioning.- Cushion inserts shall be molded of polyurethane, polyether base formulation conforming to Specification MIL-P-26514 and as specified on applicable drawings. Excessive flash of each molded insert shall be trimmed to comply with dimensions specified on applicable drawings. The cushion inserts shall be uniform texture with no hard spots in the cavity surfaces that may affect shock isolation performance. All exposed surfaces and cavities shall have a smooth skin surface, free of voids in cavities and uniform in color with no transfer of release agent on finished insert.

3.3.6 RF shielding.- RF shields shall be provided for item numbers 3, 9, 10, 13, 14, 17, 20 and 24 as referenced in 6.1.1. The shields shall be fabricated in accordance with applicable drawings for installation on referenced units by field personnel in accordance with instructions contained in "Operating Instruction Manual" prior to installing in applicable combination case.

3.3.7 Work table assemblies.- Two work table assemblies shall be fabricated in accordance with applicable drawings. The assemblies shall assure a positive means of retaining respective cases. Each table shall provide for stowage and operation of Line Regulators, items 22 and 23 as referenced in 6.1.1. A ground wire shall be provided from the suspended back which houses the line regulator to a structural member of the table. Power receptacle assemblies for "Regulated" and "Unregulated" power source, shall be wired and assembled to assure good electrical contact and continuity.

3.3.8 Van frame assembly.- The mounting frame assembly shall be fabricated in accordance with applicable drawing and shall provide a positive means of anchoring the combination cases and work tables in an M292 van.

3.4 Performance requirements.

3.4.1 Pressure retention.- Each combination case shall be capable of maintaining a constant pressure of 1.5 psi above atmosphere without loss for a period of 15 minutes.

3.4.2 Edgewise drop test.- Each combination case shall be subjected to the procedure specified in 4.5.2.3. The maximum shock transmitted to the equipment through the suspension system shall not exceed the specified shock acceleration as specified in Table I, paragraph 4.5.2.3 (applicable to pre-production cases only).

3.4.3 Vibration.- Each combination case shall be subjected to the procedure specified in 4.5.2.4. During the low frequency vibration, there shall be no natural frequency existing below seven (7) cycles per second. There shall be no mechanical failure as a result of specified excursion and resonant dwell. (applicable to preproduction cases only).

3.4.4 Road test.- The Set, defined in 3.1, shall be subjected to the procedure specified in 4.5.3. There shall be no breakage of any parts, permanent deformation of any part, loosening of threaded or other parts; other mechanical or electrical damage. In addition, there shall be no evidence of collision between parts of the set.

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3.5 Identification marking.- Identification marking, unless otherwise specified, shall be in accordance with MIL-STD-130.

3.6 Workmanship.- Workmanship shall be of the quality necessary to produce complete set free from defects which might affect their strength, serviceability and durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements.- Inspection requirements and procedures shall be in accordance with Specification MIL-I-45208.

4.3 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.4 Inspection of components.- All components shall be inspected for conformance to the requirements outlined in their respective component specifications and drawings. The inspection system required by Specification MIL-I-45208 shall contain those inspection procedures or other testing procedures equal to or better than those outlined in the detail component specifications. However, the Government reserves the right to inspect any requirement specified on the drawings or specification and to reject any individual nonconforming item.

4.5 Preproduction testing.- Prior to production, preproduction cases and set specified in paragraph 3.1 and 3.1.2 shall be inspected by the contractor under surveillance of the Government. The inspection shall be conducted at the contractor's facility or Government approved laboratory, using proven test equipment. The sample case and set shall be inspected for all requirements of this specification by utilizing examination and testing prescribed herein.

4.5.1 Preproduction failure.- Should a preproduction case or set fail to pass the prescribed tests, the contractor shall make the necessary corrections. When corrections have been made, the preproduction case or set shall be re-examined and re-tested by the contractor under the surveillance of the Government. Manufacture of production lots will not commence until an acceptable model is submitted.

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4.5.2 Preproduction cases.- Preproduction cases shall be subjected to tests in the following sequence:

(a) Pressure retention test
(b) Edgewise drop test
(c) Pressure retention test
(d) Vibration test
(e) Pressure retention test

4.5.2.1 Preparation for tests.- The containers shall be fitted with actual equipment or dummy loads of equivalent weight and mass distribution. An approved measuring device to record maximum acceleration shall be securely fastened to the dummy load or equipment. After instrumentation, the container shall be tested in the sequence given in 4.5.2.

4.5.2.2 Pressure retention test.- Each container shall have the breather valve removed and replaced by an appropriate plug. A suitable connection shall be made for the introduction of air pressure. A gage shall be connected to indicate pressure within the container. Air pressure shall be applied up to 1 1/2 pounds per square inch and the intake valve closed to retain pressure. The container shall comply with the requirements of 3.4.1.

4.5.2.3 Edgewise drop test.- Each loaded container in its horizontal position shall be supported at one end of its base on a sill or block five inches in height. The opposite end of the container shall be allowed to fall freely from the specified height indicated in Table I onto a steel, concrete or stone surface of sufficient mass to absorb the shock without deflection. The test shall be applied twice to opposite ends of the container.

TABLE I - DROP TEST

CASE NO.	HEIGHT OF DROP	MAXIMUM ACCELERATION "G's"
A-1	28.5"	24 +10%
A-3	29.5"	23 +10%
A-4	29.75"	25 +10%
B-7	26."	23 +10%

4.5.2.4 Vibration.- Each combination case shall be vibrated in three mutually perpendicular planes over the frequency ranges of 2.5 to 33 cps at a double amplitude of .06 inches and 33 to 55 cps at a double amplitude of .03 inches. A resonant search shall be made in two (2) thirty second sweeps from 2.5 to 33 and return to 2.5 cps at .06 inch double amplitude displacement, and two (2) thirty second sweeps at 33 to 55 and return to 33 cps at .03 inch double amplitude displacement. A resonant dwell shall be conducted for a ten-minute period at each of the resonant conditions found in the resonant search.

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4.5.3 Road test.- The complete preproduction set shall be installed by the contractor in an M292 van in its road transportation condition. Representative cases selected by the Government shall be fitted with their test equipment prior to road testing. The selected cases will be checked for operation by the Government prior to and immediately after the conclusion of all road testing. The preproduction set shall be transported for the specified distance over each of the following types of roads to the indicated speeds:

(a) Dirt of umimproved road, 30 miles at 10-15 mph including not less than 15 sudden stops from 10 mph

(b) Paved roads, 50 miles at 35 mph, including not less than 25 sudden stops from 25 mph.

At the conclusion of the test, the containers shall comply with the requirements of 3.4.4.

4.5.4 Mechanical and visual inspection.- The entire preproduction set shall be given a thorough mechanical and visual inspection to determine the quality of material and workmanship, and compliance with the requirements of this specification. Particular attention shall be given to the following:

(a) Completeness (including detailed check-off list)
(b) Identification markings
(c) Fit of components in their respective positions
(d) Check of mounting means and mounting hardware
(e) Cable runs between components including plugs and receptacles
(f) Grounding connections
(g) Ease of operation of adjustable and sliding parts
(h) Overall dimension check
(i) Maintenance accessibility of components
(j) Check of provisions for securing components during transit
(k) Check of shock mounting provisions
(l) Check of vent provisions
(m) Other visual defects
(n) Finishes
(o) Weld joints
(p) Accessibility of components and parts for servicing
(q) Dimensional and visual check of molded inserts
(r) Molds free of objectionable odors, voids, flash, hard spots
(s) Packaging, packing and marking

4.5.5 Electrical test.- Each power receptacle in the combination cases and work tables shall be given a continuity test to determine if wired and connected correctly and that a good electrical contact is obtained.

4.6 Production testing.

4.6.1 Pressure retention test.- Each container shall be subjected to the pressure retention test specified in 4.5.2.2, and shall comply with the requirements of 3.4.1.

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4.6.2 Mechanical and visual inspection.- Each container shall be carefully examined in accordance with 4.5.4 to determine conformance with the requirements of 3.6.

4.6.3 Electrical test.- Each power receptacle assembly shall be tested as specified in 4.5.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Unless otherwise specified, the entire set shall be packaged using normal commercial preservation methods and shall be prepared for delivery to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to point of delivery. Each combination case shall be protected with a suitable cushioning material to prevent abrasion to paint finishes. The preparation for delivery shall conform to the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment.

5.1.1 When necessary, all screws, bolts, nuts and other holddown provisions shall be inspected and tightened.

5.1.2 All loose equipment shall be secured in such a manner that movement, from rough handling or shipping conditions, shall be eliminated.

5.1.3 All container covers shall be tightly secured by provided latches.

5.2 Marking.- Marking for shipment shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- The set of combination cases with work tables as shown in Figures 1 and 2 is intended to house a complete DC, Low Frequency and Physical Calibration Set in an M292 van as a mobile calibration laboratory and to adequately protect contents during all encountered hazards of shipment and extended storage.

6.1.1 Calibration standards.- DC, Low Frequency and Physical Calibration Standards are as referenced by item and case designation listed in Table II.

TABLE II

CALIBRATION STANDARD EQUIPMENT LIST
FOR
DC, LOW FREQUENCY AND PHYSICAL CALIBRATION STANDARDS

ITEM NO.	EQUIPMENT	PART NO.	CASE NO.	DRAWER NO.
1	Voltmeter	8616477	A1	
2	Generator, AP	8616395	A1	
3	Voltmeter	7659895	A1	

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TABLE II - (Continued)

ITEM NO.	EQUIPMENT	PART NO.	CASE NO.	DRAWER NO.
4	Power Supply	8616191	A2	
*5	Power Supply	8616191	A5	
6	Microwave power meter	(HP430C)	A2	
7	Transformer, Var. 79	7907139	A2	
8	Power Supply	7907346	A3	
9	Analyzer, Distortion	7907147	A4	
10	Generator Signal	7659913	A4	
11	Calibrator, Volt	8205523	A2	
12	Voltmeter	8616091	A5	
13	Vectrolzyer AD-YU	7907235	A5	
14	Frequency Meter (Counter)	8616091	B6	
15	Converter Freq.	8616097	B6	
16	Converter Freq.	8616094	B6	
17	Comparator Freq.	8616320	B8	
18	Oscilloscope	8205525	B7	
19	Voltmeter	7907278	B7	
20	Generator, Pulse	7907148	B7	
21	Generator, Signal	8598927	B8	
22	Regulator Line	8616469	Work Table	
23	Regulator Line	8616469	Work Table	
24	Transfor, Osc.	8616100	A9	
25	Volts, Standard	8579438	A9	
26	Generator, Signal	8598968	A10	
27	Amplifier, Video	7659900	A-12	3
28	Multiplier, Period	7659902	A-12	3
29	Power Supply	8616472	A-11	3
30	Voltmeter, AC-DC	8579427	A-13	2
31	Multimeter Polyranger	841345	A-12	2
32	Voltmeter	8616317	A-13	3
33	Voltmeter, KV, DC	8616326	A-12	2
34	Milli-Ammeter, DC	7909145	A-11	3
35	Standard, One Volt	8579444	A-15	Mold
36	Millivolemeter	7907213	A-13	2
37	Resistance, Decade	8616415	A-14	1
38	Resistance, Decade	8616414	A-14	1
39	Resistance, Decade	8616452	A-11	3
40	Resistance, Decade	8616416	A-14	3
41	Inductance, Stand.	8205515	A-14	2
42	Inductance, Stand.	8205516	A-14	2
43	Inductance, Stand.	8205517	A-14	3
44	Inductance, Stand.	8205518	A-14	3
45	Inductance, Stand.	8205519	A-14	3
46	Inductance, Stand.	8205520	A-14	3
47	DC Amp Meter	7907275	A-13	3
48	Bridge, Resist.	841532	A-11	3

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TABLE II (Continued)

ITEM No.	EQUIPMENT	PART NO.	CASE NO.	DRAWER NO.
49	Capacitor, Stand.	7907233	A-16	Mold
50	Galvanometers	840594	A-17	2
51	Resistor (5 pcs.) Comparison Set	8616256	A-13	1
52	Calibrator Resist.	8598965	A-14	2
53	Calibrator Resist.	8598966	A-14	2
55	Resistor, Decade Power	7907234	A-17	1
56	Resistor, Decade Power	7907234	A-17	1
57	Gage, Compound	7907330	A-10	1
59	Gage, Vacuum	7907329	A-10	1
60	Tester, Resiliency	8169612	A-13	3
61	Gage, Block Set	7907456	A-14	1
62	Thermometer	7907119	A-13	3
63	Hydrometer Set	7907391	A-13	2
64	Adapter, L.H.	7907277	A-13	1
65	Adapter, L.H.	7907276	A-13	1
66	Adapter, Multiple	8598969	A-13	1
67	Tensiometer, Dial Indicator	8598971	A-13	3
67A	Rheostat	7909982	A-15	Mold
68	Weight, Gram Troemner	7907394	A-17	2
69	Stop Watch - 382-1/100 sec		A-13	1
70	Stop Watch - 401-1/10 sec		A-11	3
71	Voltage, Divider	7907395	A-17	2
72	Micro-Pot Set	7907350	A-14	1
73A	Tensiometer, 2 Pc.	7907286	A-18	Mold Upper Lower
74	Tester, Torque and Case	8579433 8598979	A-15	Mold
76A	Test Kit, Flow Rate	7907273	A-15	Mold
76B	Test Kit, Flow Rate	7907273	A-16	Mold
77	Ammeter, AC	7907462	A-13	2
80	Timer, Interval	7659901	A-12	3
82	Transformer, Var. Power	8205524	A-17	2
84	Dynamometer	8030812	A-16	Mold
85	Gage, Starrett	7902239	A-13	1
87	Set, Stand	66253476016	A-16	Mold
88	Thermometer	8032312	A-13	1
89	Adapters			Loose Drawer
90	Assembly, Antenna	8616486	A-17	Loose Drawer
91	Adapter, Conn.	59356292354	A-11	2
92	Adapter, Set	66253476013	A-11	2
93	Assembly, Cable	11358795416	A-11	1
94	Assembly, Resistor	7907455		Loose Drawer
95	Assembly, Resistor	7907454		Loose Drawer
97	Box, Interconnecting	11905380137	A-11	1
99	Connector, Plug	11356014754	A-11	1
100	Connector, Switch	8598928	A-11	3

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TABLE II -(Continued)

ITEM No.	EQUIPMENT	PART NO.	CASE NO.	DRAWER NO.
101	Connector, "T" Coax	8579426	A-13	1
102	Resistor Load	7907280	A-13	1
103	Mount, Univ. Bolometer	8579046	A-13	1
104	Resistor, Fixed Comp.	59052992036	A-11	1
105	Resistor, Fixed Film (4 Pcs.)	59056287749	A-11	1
106	Rheostat (2 pcs.)	7907281	A-18	Mold Upper
107	Resistor, Var. (2 Pcs.)	59058184007	A-13	1
108	Set, Stand, Aux.	11905947694	A-11	1
109	Set, Stand.	11356890563	A-11	2
110	Switch, Knife	7907120	A-11	1
111	Transformer	59501698856	A-11	1
116	Altimeter (2 Pcs.)	6610-6707511	A-12	3

NOTE: *Compartment for Item 5 will also accept alternate units, John Fluke No's. 801 and 803. Adjustment of rear brackets is required as unit depth varies.

6.1.2 Humidity indicator.- The humidity indicator is intended for long term storage or overseas shipment for cases containing and housing calibration equipment. For this application the cases shall be desiccated with proper units of desiccant determined by Formula II of Specification MIL-P-116.

6.2 Ordering data.- Procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Conditions under which Government-furnished property will be made available to the contractor.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplies the said drawings, specifications or other data is not to be regarded by implication or otherwise as in such a manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

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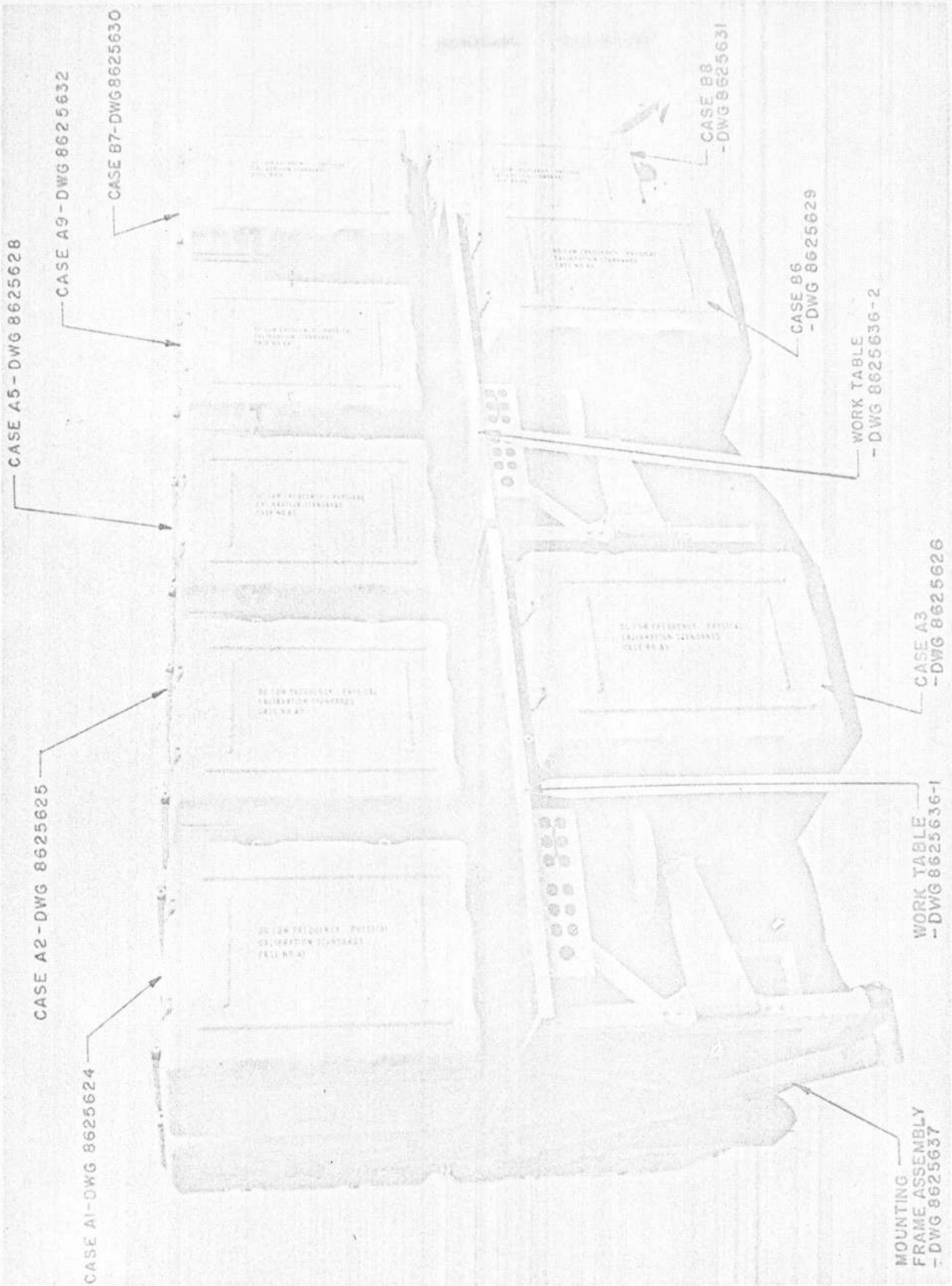


FIGURE 1. MULTIPURPOSE COMBINATION CASES W/WORK TABLES SET - DWG 8625640, WORK SIDE

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FIGURE 2. MULTIPURPOSE COMBINATION CASES W/WORK TABLES SET – DWG 8625640, STORAGE SIDE

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